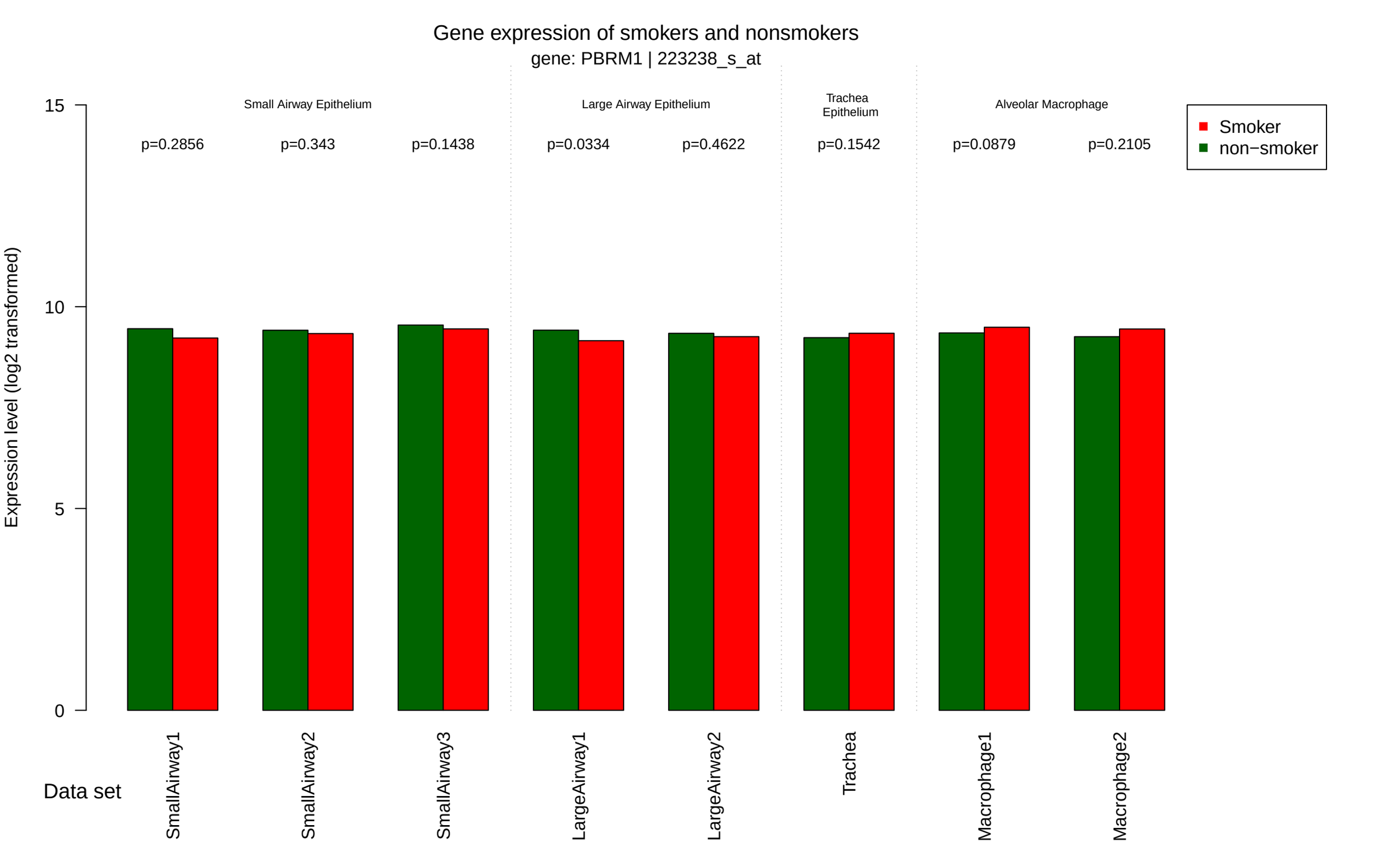
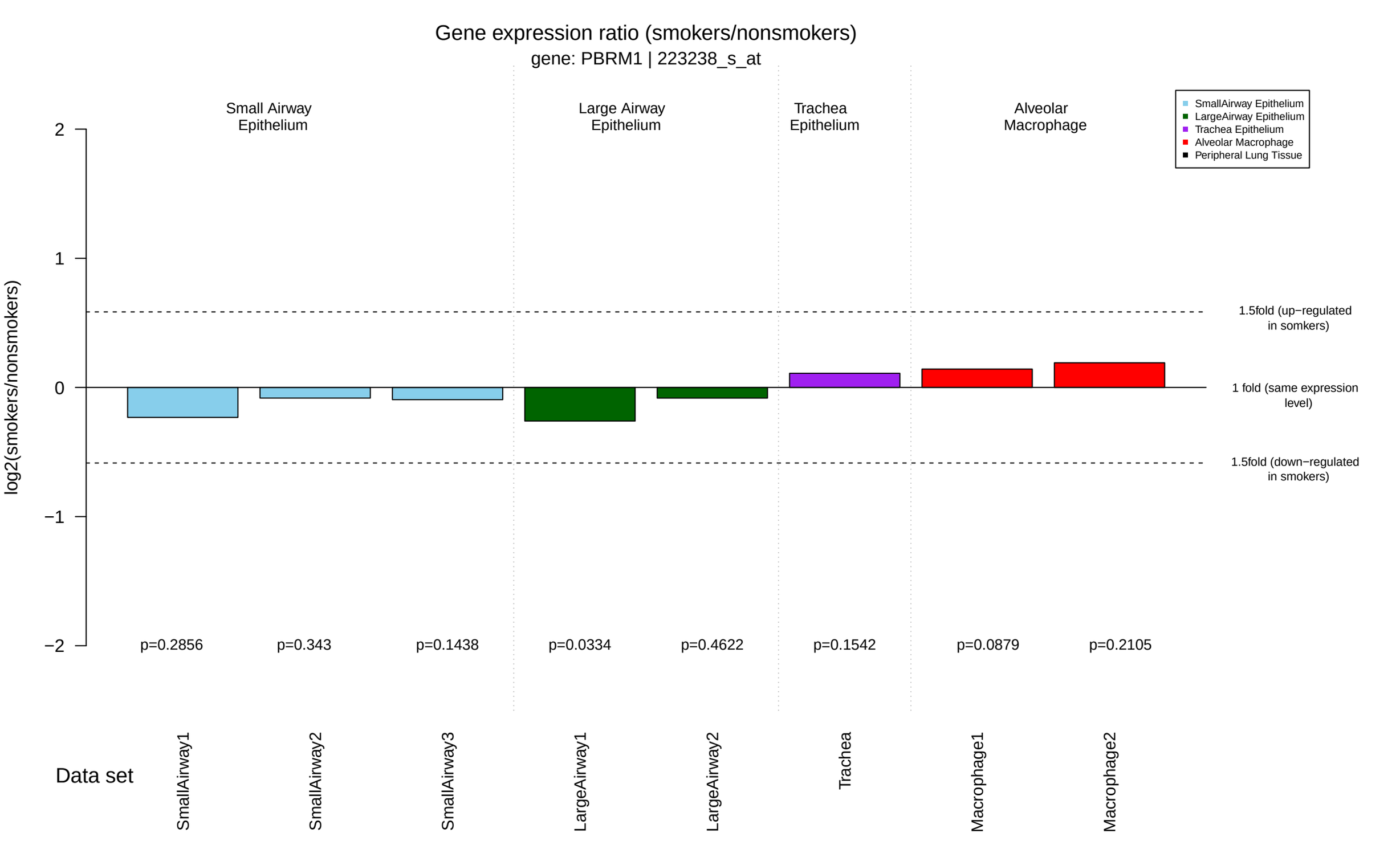
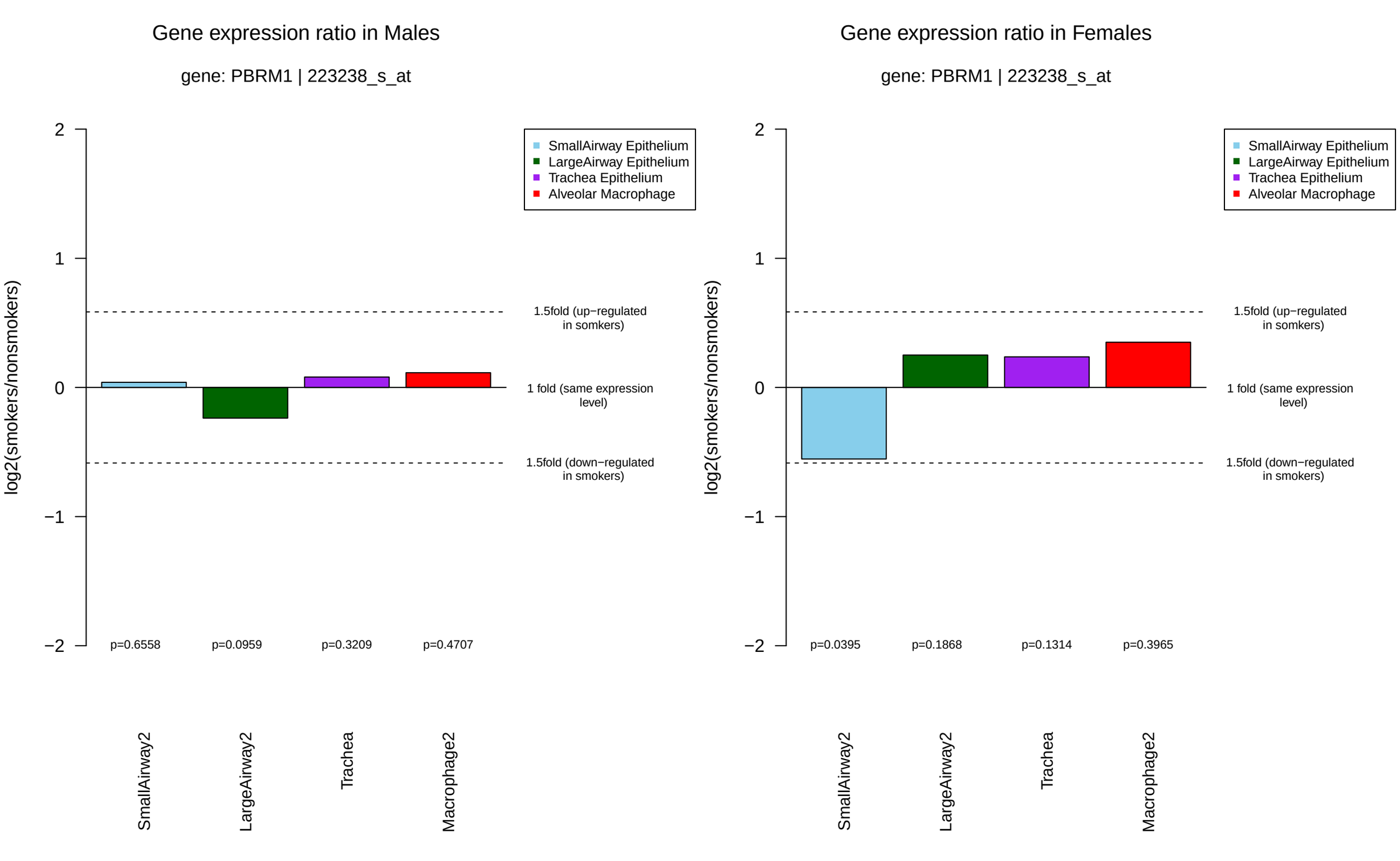
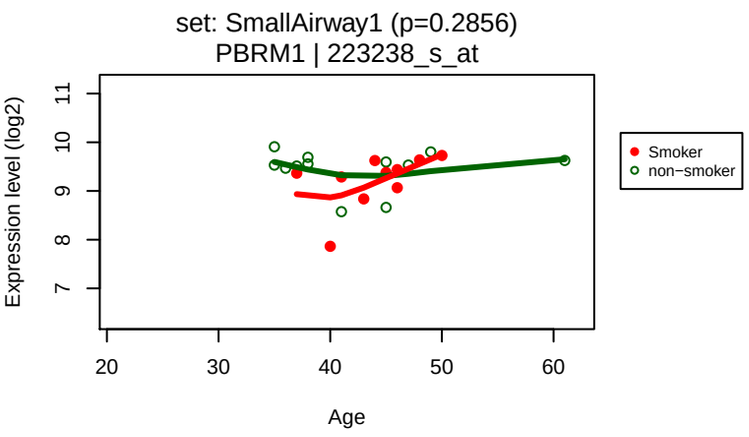
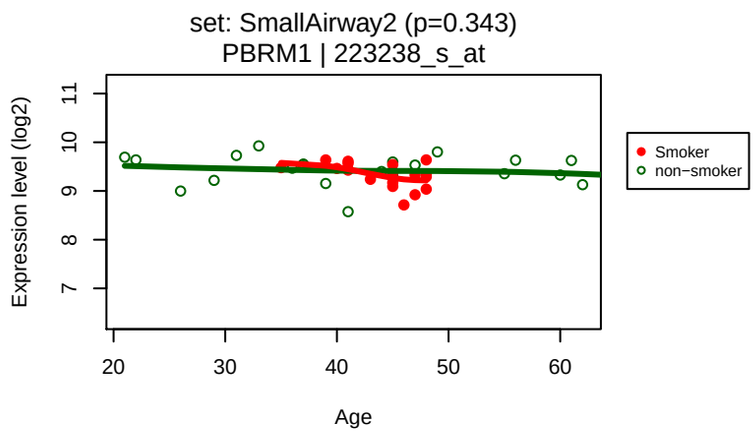
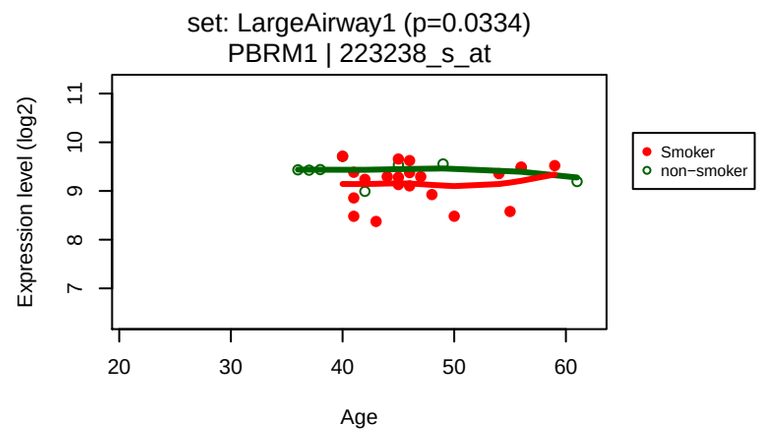

Gene expression of smokers and nonsmokers
gene: PBRM1 | 223238_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2856
p=0.343
p=0.1438
p=0.0334
p=0.4622
p=0.1542
p=0.0879
p=0.2105
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

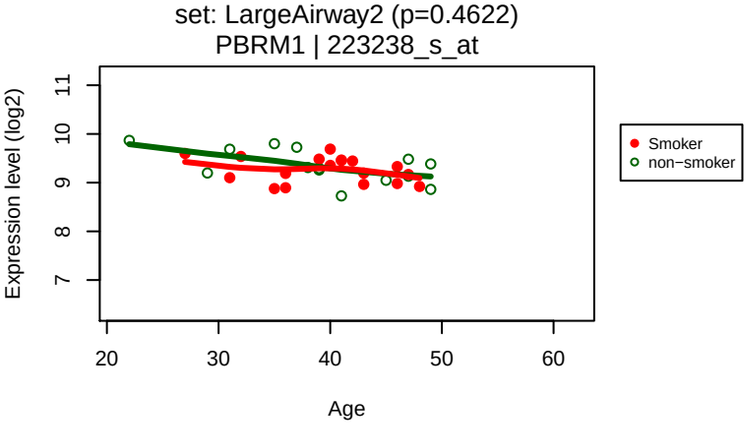

Gene expression ratio (smokers/nonsmokers)
gene: PBRM1 | 223238_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2856
p=0.343
p=0.1438
p=0.0334
p=0.4622
p=0.1542
p=0.0879
p=0.2105
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

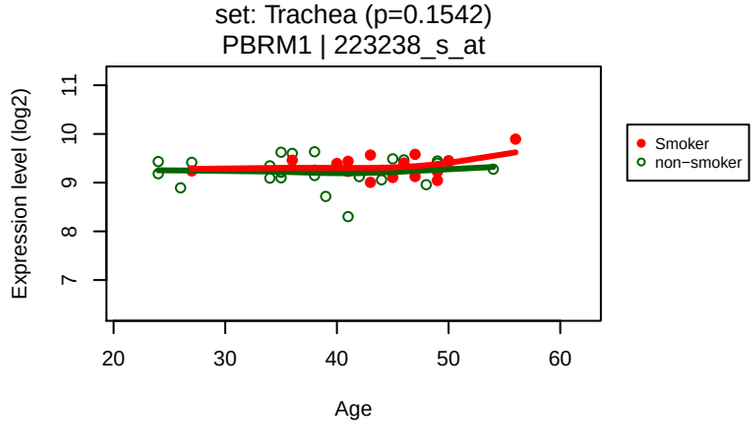

Gene expression ratio in Males
gene: PBRM1 | 223238_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6558
p=0.0959
p=0.3209
p=0.4707
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PBRM1 | 223238_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0395
p=0.1868
p=0.1314
p=0.3965
SmallAirway2
LargeAirway2
Trachea
Macrophage2

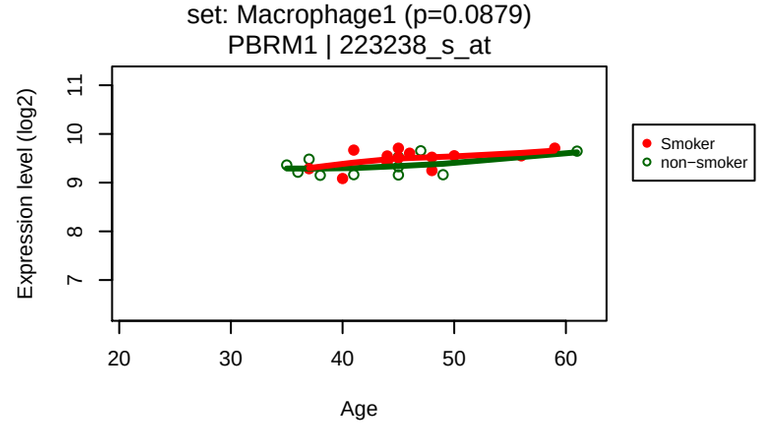

set: SmallAirway1 (p=0.2856)
PBRM1 | 223238_s_at
Expression level (log2)
Age
Smoker
non−smoker

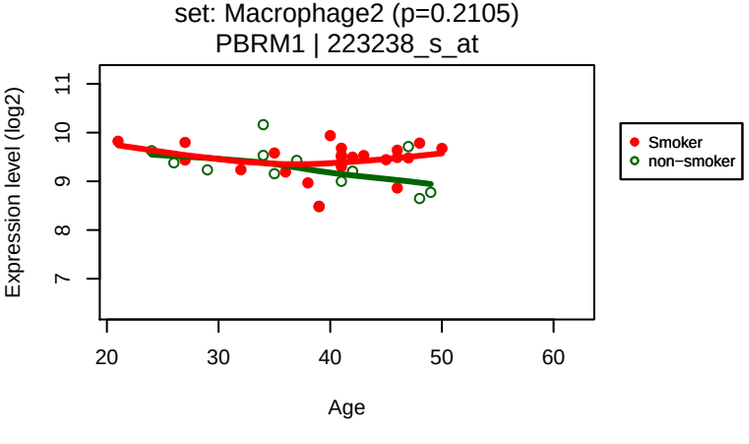

set: SmallAirway2 (p=0.343)
PBRM1 | 223238_s_at
Expression level (log2)
Age
Smoker
non−smoker

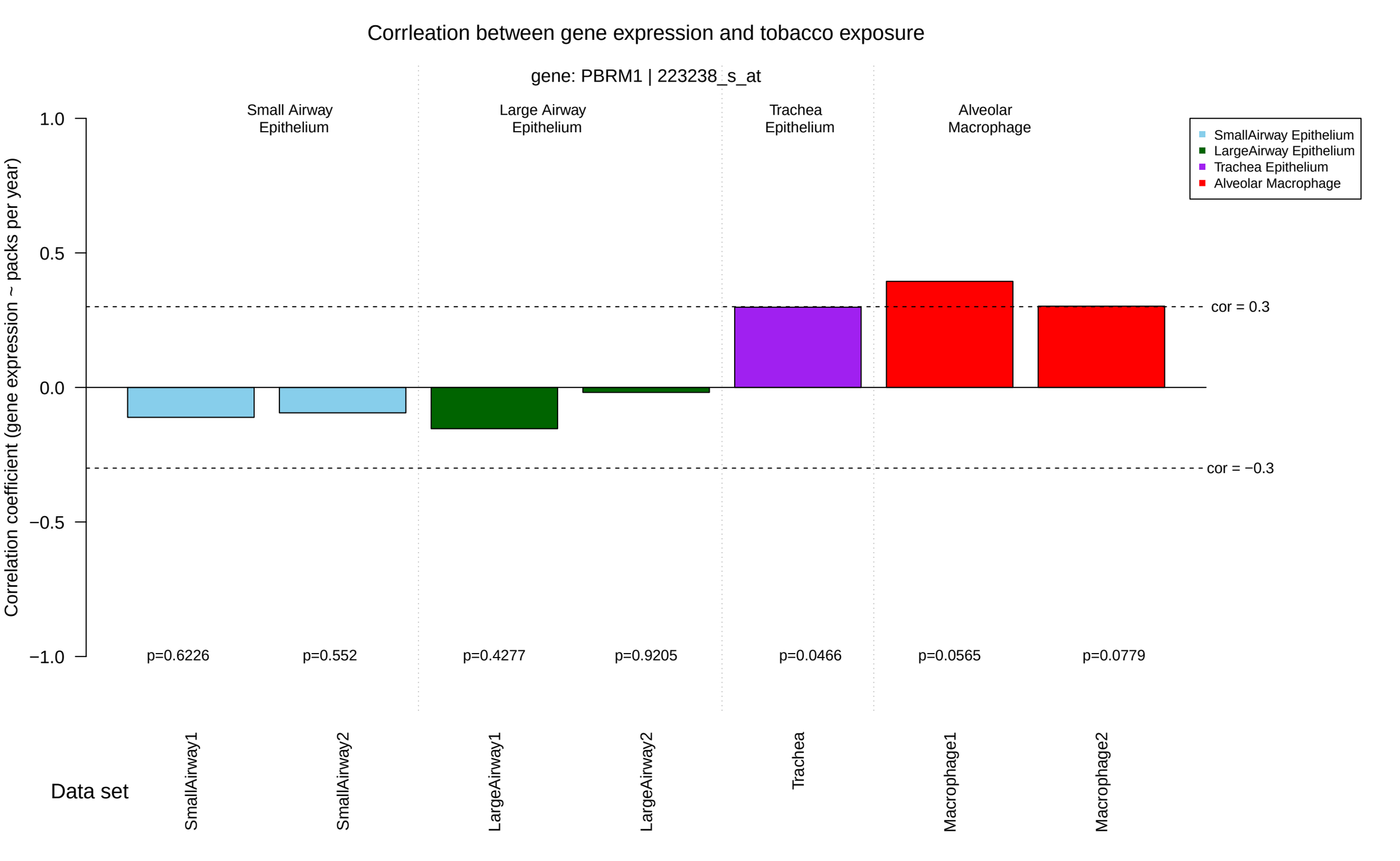

set: LargeAirway1 (p=0.0334)
PBRM1 | 223238_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4622)
PBRM1 | 223238_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1542)
PBRM1 | 223238_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0879)
PBRM1 | 223238_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2105)
PBRM1 | 223238_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PBRM1 | 223238_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6226
p=0.552
p=0.4277
p=0.9205
p=0.0466
p=0.0565
p=0.0779
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2